SYNTHETIC DEMO DATA

SkyStay

From flights to hotel guests: a scenario simulator for Abu Dhabi

A personal project by Daniel

The idea

Abu Dhabi plans flights and hotels with different numbers. SkyStay connects air capacity to hotel demand, so a planner can ask "what if" and get an answer with honest ranges.

A working prototype: data pipeline, three model layers, back-testing with calibrated uncertainty, a scenario engine with 11 levers, an 8-page simulator app, and a plain-language planner brief. Reproducible with one command:

```
skystay all
```

Shown on synthetic demo data with known ground truth: the method is demonstrated, not real Abu Dhabi figures.

THE PROBLEM

Flights and hotels are planned with different numbers, and nothing links them

- **Only some arriving passengers become hotel guests.** Many are transferring onward, returning residents, or staying with friends and family.
- **The two datasets disagree on "country".** Flights record the *departure country*; hotels record the guest's *nationality*. A London flight carries many nationalities.
- **Separate teams, separate numbers.** Route development and hotel planning have no shared model, so capacity decisions are made without knowing what they do to hotel demand.
- **The share that converts changes** by source market and by season, so a single ratio is wrong.

What planners cannot answer today

How many hotel guests would a new route bring, from a country with no service today, and in which months?

What do hotels lose if a top market cuts winter frequency by 30%?

Which markets fill the most hotel rooms per seat, so incentives go where they matter?

How confident can we be, and which assumption drives the answer?

Scheduled seats × load factor **Passengers** − transfer / transit **Visitors by nationality** × hotel capture + non-air guests **Hotel guests** × length of stay **Guest nights**

DATA AND THE COUNTRY CATCH

Departure country ≠ nationality: the back-test chooses how to reconcile them

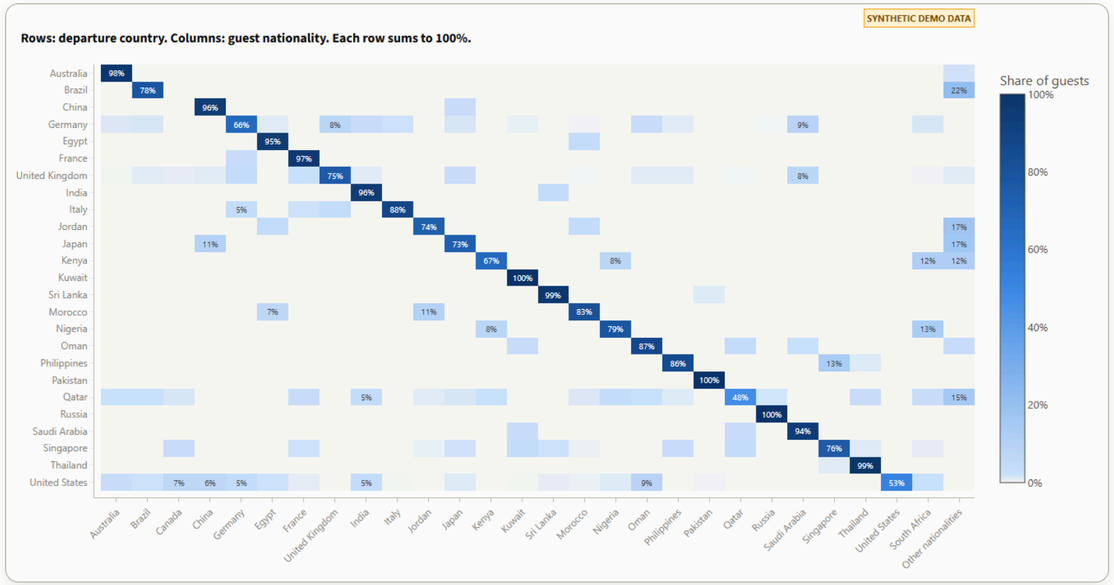
Input (2019-01 to 2025-12, 84 months)	Grain
Flight schedule	route × day · 25 countries, 37 airports
Passenger traffic	airport × month · arrivals, transfers
Hotel performance	nationality × month · 29 nationalities
Calendars	56 events · holidays · seasonality

Quality checks flagged 8 records (missing months, duplicates, load factor above 100%): all logged, none dropped silently.

Reconciliation level	Chain CV	Hybrid CV	Hybrid holdout
L0 identity	6.5%	5.9%	6.6%
L1 regional	7.8%	7.2%	7.1%
L2 learned bridge	**6.2%**	**5.7%**	**6.5%**

- **L0:** origin country = nationality (naive). **L1:** both sides grouped into 11 regions.
- **L2:** a learned non-negative origin → nationality matrix, ridge-pulled toward a prior, plausible pairs only (same region + hubs), year-block bootstrap intervals.

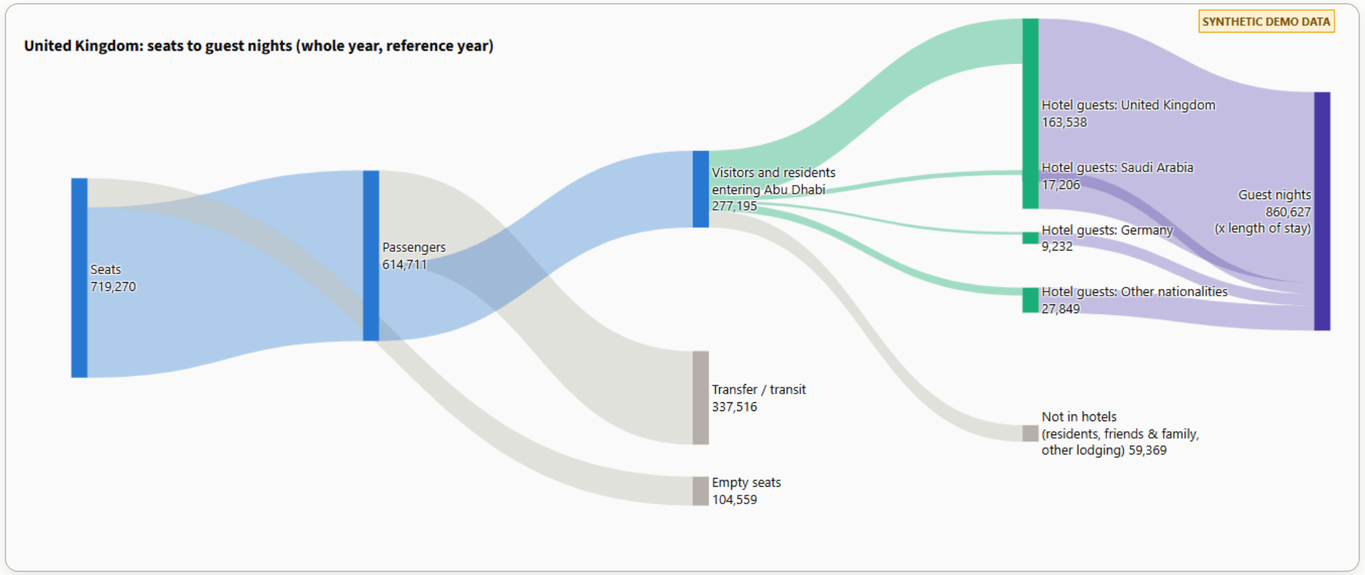
The back-test chose L2 learned bridge (WMAPE on monthly hotel guests by market).



Learned bridge: share of each origin's hotel guests by nationality (rows sum to 100%).

THE SOLUTION

A transparent conversion chain: every step visible, sourced and adjustable



The simulator's Conversion chain page: United Kingdom flights, reference year. Hotel guests include other nationalities carried by the route (the learned bridge).

Assumptions for United Kingdom (estimated from data)

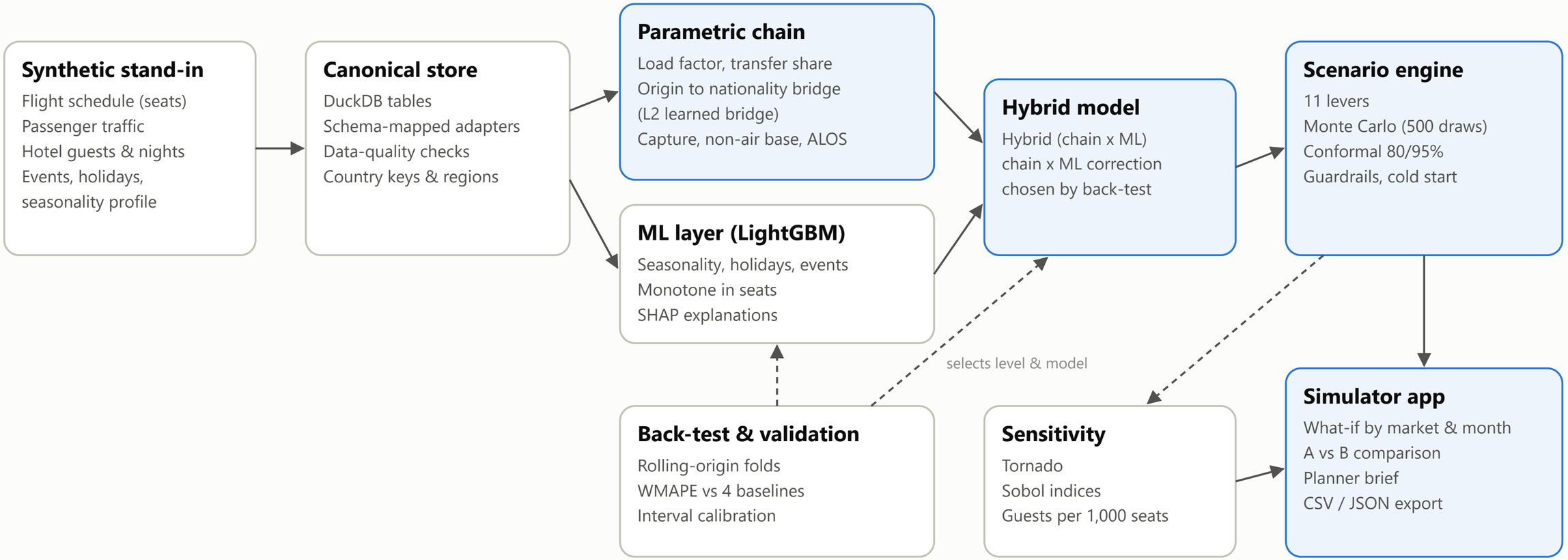
Step	Value	80% interval
Load factor	85.2%	0.843 to 0.861
Transfer / transit share	54.8%	0.539 to 0.558
Hotel capture (United Kingdom nationals)	0.79	0.71 to 0.88
Length of stay (nights)	4.16	4.11 to 4.22
Non-air share of guests	28%	not a lever

Every assumption sits in **config/assumptions.yaml** with value, unit, 80% interval, n_obs, method, source and an adjustable flag. Empirical-Bayes shrinkage by month; thin and new markets borrow from their region (*cold start*).

Non-air demand (UAE residents, land arrivals, other UAE airports) is estimated from data: 45% of guests in the latest year. Flight levers never move it.

SYSTEM DESIGN

From raw aggregates to a what-if answer in one pipeline



Stack: Python, DuckDB, statsmodels, LightGBM, SHAP, SALib, Streamlit, Plotly, Playwright. Runs on a laptop.

Speed: a 12-month scenario with 500 Monte Carlo draws runs in about 70 ms, so every slider move updates instantly.

Guarantees: 65 automated tests, including no leakage, parameter recovery, monotone response to seats and bridge locality.

SYNTHETIC DEMO DATA

MODEL AND VALIDATION (ON SYNTHETIC DATA WITH KNOWN GROUND TRUTH)

Back-tested on held-out years: the hybrid beats every baseline

Holdout WMAPE, hotel guests

6.5%

monthly, by market · seasonal naive 8.9%

Holdout WMAPE, guest nights

6.9%

bias -1.1% on guests

Interval coverage

79% / 91%

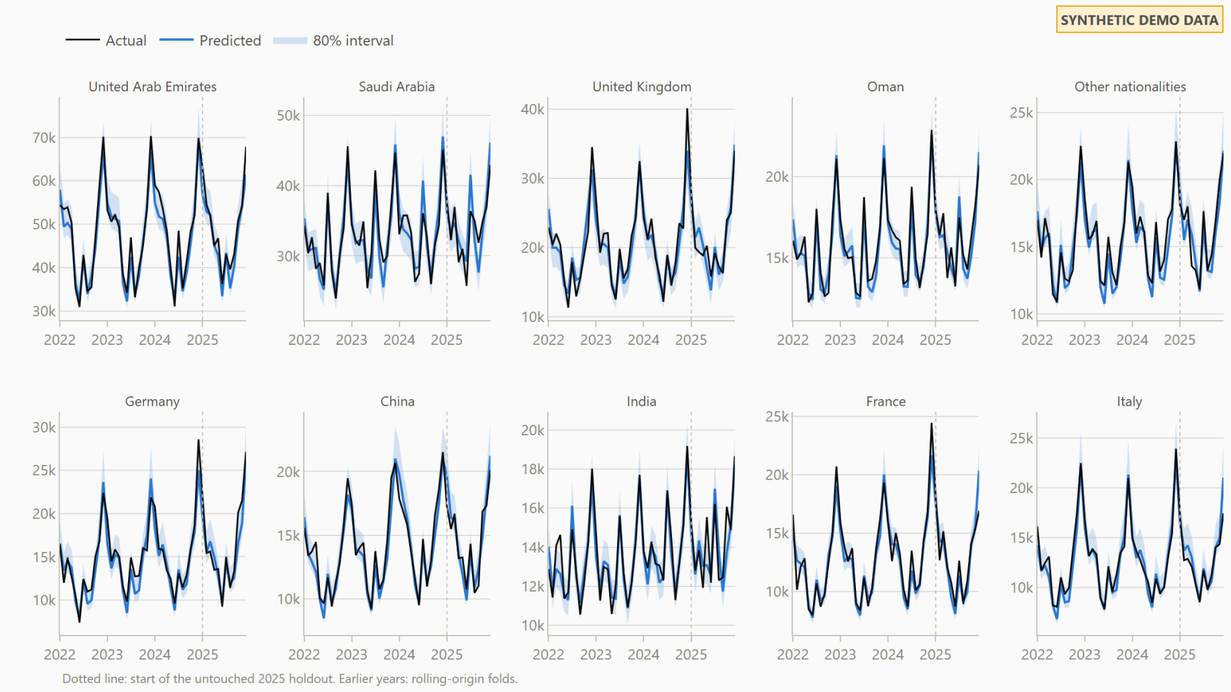
of holdout months inside the 80% / 95% ranges

Parameter recovery

7/7

known synthetic parameters recovered within tolerance

Hotel guests: actual vs back-test predictions, top 10 markets (on synthetic data with known ground truth)

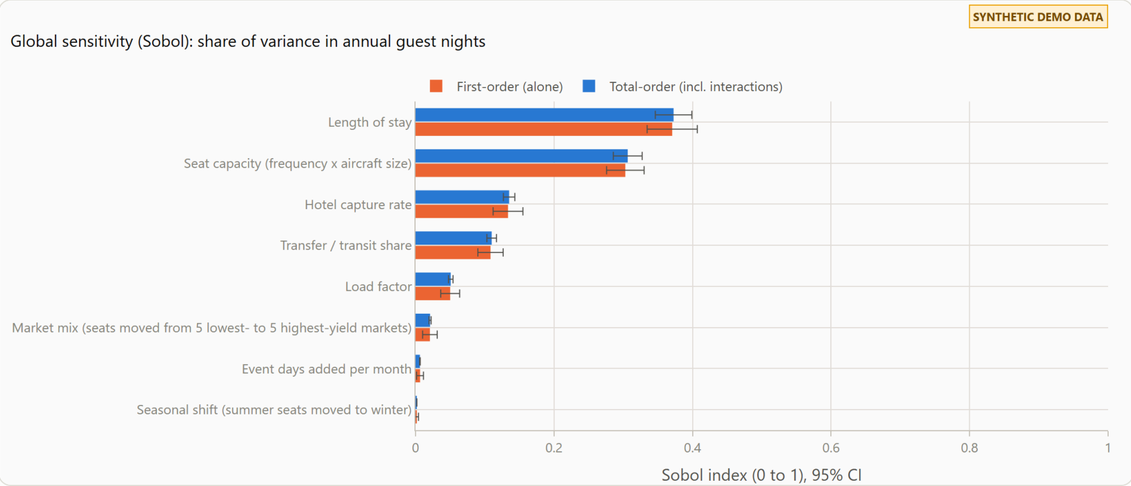


Model (L2 learned bridge)	CV	Holdout
Seasonal naive	8.3%	8.9%
Parametric chain	6.2%	6.7%
Econometric	7.8%	8.6%
ML only	7.1%	6.9%
Hybrid (chain x ML)	**5.7%**	**6.5%**

WMAPE = Σ|actual − predicted| / Σ actual, monthly hotel guests by market. Three rolling-origin folds (test 2022, 2023, 2024), refit on pre-cutoff data only; 2025 scored once. Intervals: split-conformal on out-of-fold residuals.

SENSITIVITY

What moves hotel demand most, and which routes fill the most rooms



Global (Sobol, 20,480 model runs): length of stay explains 0.37 of the variance in annual guest nights, then seat capacity (frequency x aircraft size) (0.31) and hotel capture rate (0.14). A one-at-a-time tornado (±10% and ±1 SD) is in the app.

Hotel guests per 1,000 added seats

Market	Guests / 1,000 seats	80% range
United States	558	537 to 583
China	463	438 to 521
Italy	462	437 to 521
Japan	446	361 to 542
France	414	383 to 469
… lowest yield		
India	87	76 to 97
Pakistan	73	63 to 83

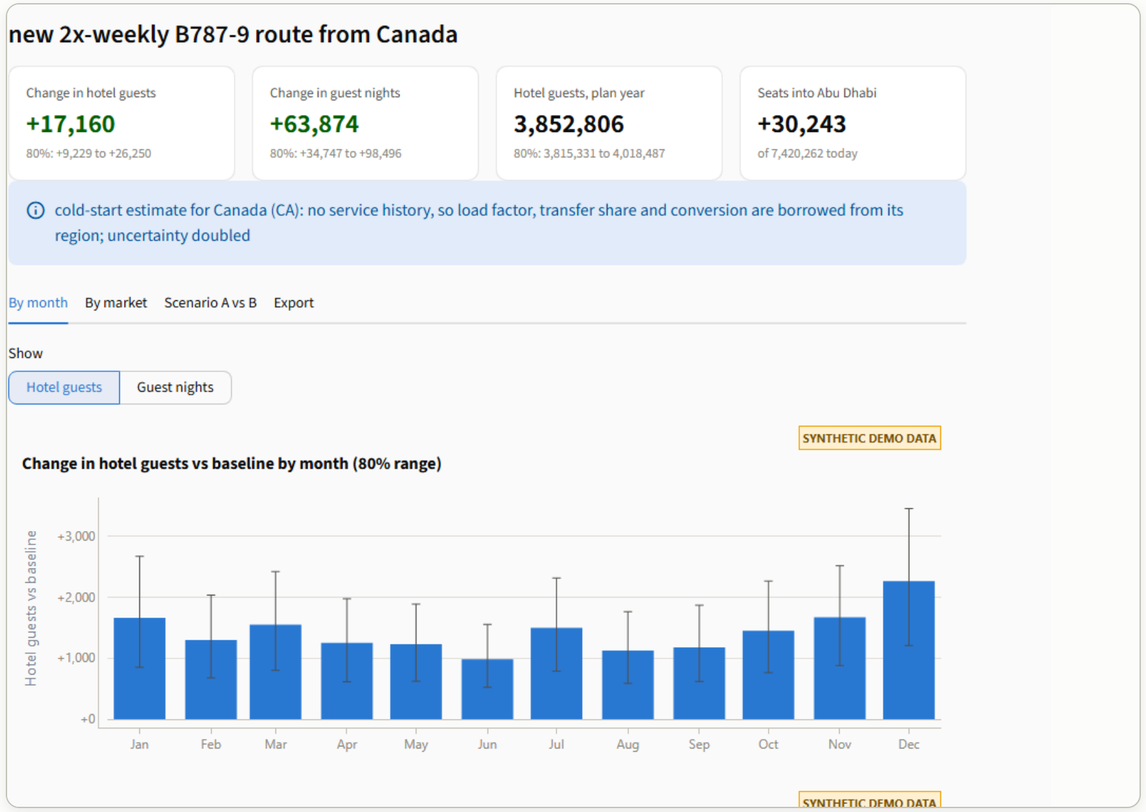
New markets (cold-start estimates): Canada 564 (317 to 842); South Korea 453 (234 to 816); Spain 362 (165 to 661).

Seasonal vulnerability: a 10% capacity cut costs most in Summer low (Jun-Aug): 159,547 guest nights (6.3% of the season).

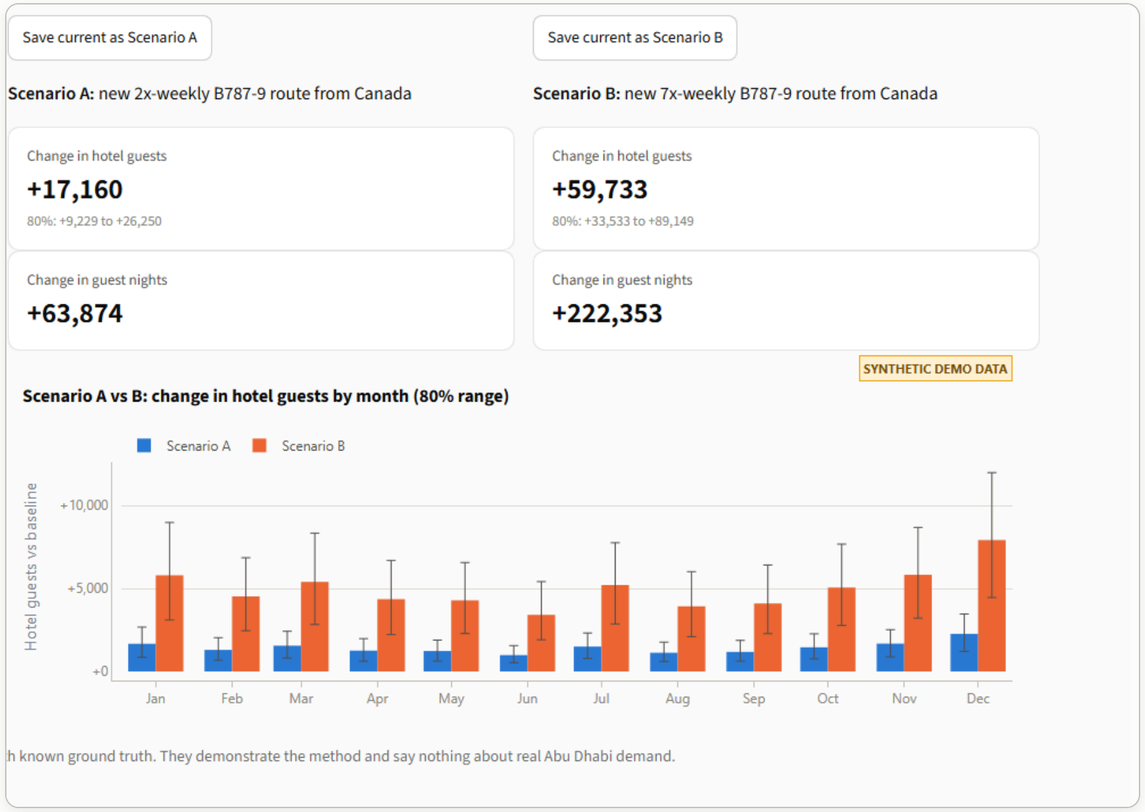
SYNTHETIC DEMO DATA

WORKING PROTOTYPE

Change a lever, see hotel guests and nights move, with honest ranges



Scenario simulator. New 2x-weekly route from Canada: +17,160 hotel guests a year (80%: +9,229 to +26,250), by month with ranges, flagged as a cold-start estimate.



Scenario A vs B. The same route at 2 vs 7 flights a week, side by side and month by month with 80% ranges; results export as CSV and the scenario as JSON.

Also in the app: winter frequency cut on United Kingdom (−30%): -20,849 hotel guests; guardrails reject impossible inputs (load factor above 100%) and warn when a lever leaves the historical range.

IMPACT AND REAL-LIFE USE PLAN

One shared number for airports, airlines and hotel planners

Who uses it

DCT tourism planning and route-development teams, with Abu Dhabi Airports and partner airlines working from the same numbers.

Decisions it changes

Which routes to incentivise, how much capacity to protect each season, where to aim source-market campaigns, what hotel demand to staff for.

Deployment path

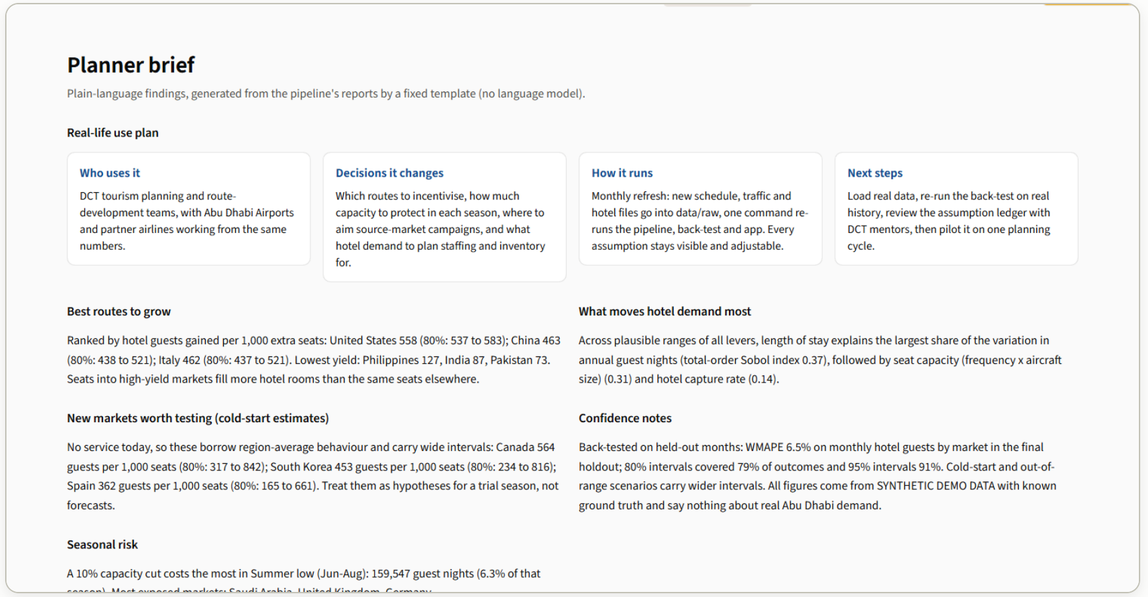
Load real data → re-run the back-test on real history → monthly refresh with one command; Docker image for internal hosting.

Next steps

Run on real tourism data, review the assumption ledger with domain experts, add hotel-capacity and price signals, then pilot on one planning cycle.

What a planner should do differently

- Rank route-development incentives by hotel guests per 1,000 seats, not by seats or passengers.
- Protect Summer low (Jun-Aug) capacity on Saudi Arabia, United Kingdom first: that is where a capacity loss hurts hotels most.
- Plan non-air demand separately: about 45% of hotel guests in the latest year do not depend on flights into Abu Dhabi, so airline deals cannot move them.
- Agree one shared number with airports and airlines: use the simulator's central estimate for targets and its 80% range for staffing and inventory buffers.
- Test new markets with a trial season and re-estimate after 6 to 12 months of data.



Planner brief

Plain-language findings, generated from the pipeline's reports by a fixed template (no language model).

Real-life use plan

Who uses it

DCT tourism planning and route-development teams, with Abu Dhabi Airports and partner airlines working from the same numbers.

Decisions it changes

Which routes to incentivise, how much capacity to protect in each season, where to aim source-market campaigns, and what hotel demand to plan staffing and inventory for.

How it runs

Monthly refresh: new schedule, traffic and hotel files go into data/raw, one command re-runs the pipeline, back-test and app. Every assumption stays visible and adjustable.

Next steps

Load real data, re-run the back-test on real history, review the assumption ledger with DCT mentors, then pilot it on one planning cycle.

Best routes to grow

Ranked by hotel guests gained per 1,000 extra seats: United States 558 (80%: 537 to 583); China 463 (80%: 438 to 521); Italy 462 (80%: 437 to 521). Lowest yield: Philippines 127, India 87, Pakistan 73. Seats into high-yield markets fill more hotel rooms than the same seats elsewhere.

What moves hotel demand most

Across plausible ranges of all levers, length of stay explains the largest share of the variation in annual guest nights (total-order Sobol index 0.37), followed by seat capacity (frequency x aircraft size) (0.31) and hotel capture rate (0.14).

New markets worth testing (cold-start estimates)

No service today, so these borrow region-average behaviour and carry wide intervals: Canada 564 guests per 1,000 seats (80%: 317 to 842); South Korea 453 guests per 1,000 seats (80%: 234 to 816); Spain 362 guests per 1,000 seats (80%: 165 to 661). Treat them as hypotheses for a trial season, not forecasts.

Confidence notes

Back-tested on held-out months: WMAPE 6.5% on monthly hotel guests by market in the final holdout; 80% intervals covered 79% of outcomes and 95% intervals 91%. Cold-start and out-of-range scenarios carry wider intervals. All figures come from SYNTHETIC DEMO DATA with known ground truth and say nothing about real Abu Dhabi demand.

Seasonal risk

A 10% capacity cut costs the most in Summer low (Jun-Aug): 159,547 guest nights (6.3% of that

Planner brief page: plain-language findings generated from the reports by a fixed template.

FEASIBILITY, LIMITATIONS AND REPRODUCIBILITY

Built to be checked: honest limits, one command, clear next steps

Limitations (full list: docs/LIMITATIONS.md)

- Synthetic data until real data is available: accuracy on real data is unknown.
- Air vs non-air demand is only weakly separable (median non-air error 0.10 on synthetic truth).
- Returning residents are folded into capture; new routes are cold-start estimates without diversion.
- Intervals calibrated on three folds: 95% range covered 91% of holdout months.

Reproducibility

- `skystay all` rebuilds data, back-test, model, sensitivity, video and deck (seed 42).
- Swap in real data: edit config/schema_map.yaml, re-run; the synthetic badge disappears.
- 65 tests: leakage, parameter recovery, golden scenarios, app smoke tests. Docker image included.

Feasibility

- Uses aggregated data only; no personal data.
- Open-source stack, runs on a laptop; scenario answers in about 70 ms.
- Every number in this deck and the video is generated from the pipeline's reports.

What's next

- Real data: map the files in config/schema_map.yaml and re-run the back-test.
- Model diversion from hub routes when a direct route opens; add hotel capacity and price.
- Longer calibration history for the intervals; a recorded human voiceover for the demo.